

GENERALIZED BIVARIATE RESIDUAL ENTROPY FUNCTION AND SOME CHARACTERIZATION RESULTS

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Summary: Shannon's entropy has been extensively used in literature as a measure of information and finds a lot of applications in several branches of science and engineering. The residual entropy function, which is the Shannon's entropy associated with the residual life, is viewed as a dynamic measure of uncertainty in dealing with life distributions. The present paper addresses the question of extending the definition of the generalized residual entropy, considered by Nanda and Paul (2006) to higher dimensions. We also consider the problem of characterizing some bivariate models using the functional form of the bivariate residual entropy function.

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1. Introduction

In the recent past, many researchers have taken keen interest in the measurement of uncertainty associated with a probability distribution. Of particular interest is the notion of entropy, introduced by Shannon (1948). If X is a non-negative random variable admitting an absolutely continuous distribution function $F(x)$ with probability density function $f(x)$, Shannon's entropy associated with the random variable X is defined as

$$H(X) = H(f) = - \int_0^{\infty} f(x) \log f(x) dx. \quad (1.1)$$

$H(f)$ is interpreted as a quantitative measure of the extent of uniformity about the random phenomena described by the probability density function $f(x)$.

Ebrahimi and Pellerey (1995) followed by Ebrahimi (1996) introduced and extensively studied the concept of residual entropy. For a non-negative random variable X , representing the life time of a component, the residual entropy function is Shannon's entropy associated with the random variable $X - t$ truncated at $t \geq 0$ and is defined as

$$H(f, t) = - \int_t^{\infty} \frac{f(x)}{\bar{F}(t)} \log \frac{f(x)}{\bar{F}(t)} dx, \quad (1.2)$$

where $\bar{F}(t) = P(X > t)$ denotes the survival function of X . $H(f, t)$ can also be written as

$$H(f, t) = 1 - \frac{1}{\bar{F}(t)} \int_t^{\infty} f(x) \log h(x) dx, \quad \bar{F}(t) > 0,$$

$h(x) = \frac{f(x)}{\bar{F}(x)}$ is the hazard function of X . Navaro et al. (2004) established that if $H(f, t)$ is increasing in t then $H(f, t)$ determines the distribution

uniquely. Given that an item has survived up to time t , $H(f, t)$ measures the uncertainty about its remaining life. For a discussion of the properties and characterisation results using the notion of residual entropy, we refer to Nair and Rajesh (1998), Sankaran and Gupta (1999) and Asadi and Ebrahimi (2000).

In studying the reliability aspects of a multi-component system with each component having a lifetime depending on the lifetimes of the next, a multivariate life distribution is employed. Reliability characteristics in the univariate case extends to the corresponding multivariate version. Even though a lot of interest has been evoked on the residual entropy function in the univariate case, only very little work seems to have been done in higher dimensions. For details see Ahmed and Gokhale (1989), Darbellay and Vajda (2000), Rajesh and Nair (2000), Nadarajah and Zografos (2005), Nanda and Paul (2005) and Ebrahimi et al. (2007). In the present paper we give an alternative definition for the residual entropy function in the bivariate set-up, and study its properties.

In a recent work, Nanda and Paul (2006) have generalized (1.2), for a unit surviving up to an age t and have provided two alternative definitions for the residual entropy. The first kind of residual entropy is defined as

$$H_1^\beta(X; t) = \frac{1}{1-\beta} \left[1 - \int_t^\infty \left(\frac{f(x)}{\bar{F}(t)} \right)^\beta dx \right], \quad (1.3)$$

for some fixed $\beta > 0$, $\beta \neq 1$ and $t > 0$ and the second kind of residual entropy is

$$H_2^\beta(X; t) = \frac{1}{1-\beta} \ln \int_t^\infty \left(\frac{f(x)}{\bar{F}(t)} \right)^\beta dx. \quad (1.4)$$

It can be noted that as the order of entropy $\beta \rightarrow 1$ in (1.3) or in (1.4), they reduce to (1.2). It is also established that under some conditions the β -residual entropy determines the distribution uniquely. In the present work we extend the

residual entropies defined in (1.3) and (1.4) to higher dimensions analogous to the extension of failure rate given in Johnson and Kotz (1975).

The rest of the paper is organized as follows. In Section 2, we present the bivariate extension of generalized residual entropies defined in (1.3) and (1.4). Characterization results for some common bivariate lifetime models using the functional form of the bivariate residual entropy function are discussed in Section 3. In Section 4 we illustrate the computation of the residual entropy function using a numerical example. Finally, Section 5 gives a brief discussion of possible extensions of the concept to higher dimensions.

2. Vector-valued residual entropy function

Let $X = (X_1, X_2)$ represent a non-negative random vector admitting an absolutely continuous distribution with density function $f(x_1, x_2)$ and survival function $F(x_1, x_2)$. Let $Y_1 = X_1|X_2 > t_2$ correspond to the random variable X_1 subject to the condition that X_2 is greater than the threshold t_2 with density function $f(x_1|X_2 > t_2)$ and survival function $\bar{F}(x_1|X_2 > t_2)$.

Observing that

$$\bar{F}(t_1, t_2) = \bar{F}(t_1|X_2 > t_2) P(X_2 > t_2), \quad (2.1)$$

we have

$$\frac{\partial}{\partial t_1} \bar{F}(t_1, t_2) = -f(t_1|X_2 > t_2) P(X_2 > t_2). \quad (2.2)$$

This gives

$$f(t_1|X_2 > t_2) = -\frac{1}{P(X_2 > t_2)} \frac{\partial}{\partial t_1} \bar{F}(t_1, t_2). \quad (2.3)$$

We define the first kind of residual entropy of order β for the random variable Y_i as

$$H_{11}^\beta(X, t_1, t_2) = \frac{1}{1-\beta} \left(1 - \int_{t_1}^{\infty} \left(\frac{f(x_1|X_2 > t_2)}{\bar{F}(t_1|X_2 > t_2)} \right)^\beta dx_1 \right). \quad (2.4)$$

(2.4) can also be written as

$$H_{11}^{\beta}(X, t_1, t_2) = \frac{1}{1-\beta} \left(1 - \int_{t_1}^{\infty} \left(\frac{-\frac{\partial}{\partial x_1} \bar{F}(x_1, t_2)}{\bar{F}(t_1, t_2)} \right)^{\beta} dx_1 \right). \quad (2.5)$$

Similarly the first kind of residual entropy of order β for the random variable $Y_2 = X_2 | X_1 > t_1$

$$H_{12}^{\beta}(X, t_1, t_2) = \frac{1}{1-\beta} \left(1 - \int_{t_2}^{\infty} \left(\frac{f(x_2 | X_1 > t_1)}{\bar{F}(t_2 | X_1 > t_1)} \right)^{\beta} dx_2 \right), \quad (2.6)$$

simplifies to

$$H_{12}^{\beta}(X, t_1, t_2) = \frac{1}{1-\beta} \left(1 - \int_{t_2}^{\infty} \left(\frac{-\frac{\partial}{\partial x_2} \bar{F}(t_1, x_2)}{\bar{F}(t_1, t_2)} \right)^{\beta} dx_2 \right). \quad (2.7)$$

For a bivariate random vector $X = (X_1, X_2)$ admitting an absolutely continuous distribution function, Johnson and Kotz (1975) define the bivariate failure rate as

$$h(t_1, t_2) = (h_1(t_1, t_2), h_2(t_1, t_2))$$

where $h_j(t_1, t_2)$ is the hazard rate of the conditional distribution of X_j given $X_{3-j} > t_{3-j}$, $j = 1, 2$. Analogous to the definition of the vector-valued failure rate, we give an alternative definition for the first kind of residual entropy of order β .

Definition 2.1

For a non-negative random vector $X = (X_1, X_2)$ admitting an absolutely continuous distribution function, the bivariate vector-valued residual entropy is defined as the vector

$$H_1^\beta(X, t_1, t_2) = \left(H_{11}^\beta(X, t_1, t_2), H_{12}^\beta(X, t_1, t_2) \right) \quad (2.8)$$

where $H_{11}^\beta(X, t_1, t_2)$ and $H_{12}^\beta(X, t_1, t_2)$ are given by (2.5) and (2.7).

We can define a second kind of residual entropy of order β for the random variable Y_1 as

$$H_{21}^\beta(X, t_1, t_2) = \frac{1}{1-\beta} \log \left(\int_{t_1}^{\infty} \left(\frac{f(x_1|X_2 > t_2)}{\bar{F}(t_1|X_2 > t_2)} \right)^\beta dx_1 \right). \quad (2.9)$$

(2.9) can also be written as

$$H_{21}^\beta(X, t_1, t_2) = \frac{1}{1-\beta} \log \left(\int_{t_1}^{\infty} \left(\frac{-\frac{\partial}{\partial x_1} \bar{F}(x_1, t_2)}{\bar{F}(t_1, t_2)} \right)^\beta dx_1 \right). \quad (2.10)$$

Similarly, the second kind of residual entropy of order β for the random variable $Y_2 = X_2|X_1 > t_1$ takes the form

$$H_{22}^\beta(X, t_1, t_2) = \frac{1}{1-\beta} \log \left(\int_{t_2}^{\infty} \left(\frac{f(x_2|X_1 > t_1)}{\bar{F}(t_2|X_1 > t_1)} \right)^\beta dx_2 \right) \quad (2.11)$$

which can also be written as

$$H_{22}^\beta(X, t_1, t_2) = \frac{1}{1-\beta} \log \left(\int_{t_2}^{\infty} \left(\frac{-\frac{\partial}{\partial x_2} \bar{F}(t_1, x_2)}{\bar{F}(t_1, t_2)} \right)^\beta dx_2 \right) \quad (2.12)$$

Definition 2.2

For a non-negative random vector $X = (X_1, X_2)$, admitting an absolutely continuous distribution function, the bivariate vector-valued residual entropy is defined as the vector

$$H_2^\beta(X, t_1, t_2) = \left(H_{21}^\beta(X, t_1, t_2), H_{22}^\beta(X, t_1, t_2) \right) \quad (2.13)$$

where $H_{21}^\beta(X, t_1, t_2)$ and $H_{22}^\beta(X, t_1, t_2)$ are given by (2.10) and (2.12).

3. Characterization results

In this section, we consider the problem of characterizing some bivariate distributions using the functional form of the residual entropy function defined in (2.8) and (2.13). First we examine whether $H_1^\beta(X, t_1, t_2)$ and $H_2^\beta(X, t_1, t_2)$ determines the distribution function uniquely. The answer is affirmative as is evident from the following theorem.

Theorem 3.1 Let $X = (X_1, X_2)$ be a non-negative random vector admitting an absolutely continuous distribution function with respect to a Lebesgue measure and with vector-valued residual entropy function $H_i^\beta(X, t_1, t_2)$ which is increasing in $t_i, i = 1, 2$. Then

1. $H_1^\beta(X, t_1, t_2)$ uniquely determines $\bar{F}(t_1, t_2)$.
2. $H_2^\beta(X, t_1, t_2)$ uniquely determines $\bar{F}(t_1, t_2)$.

Proof From (2.5), we have

$$\int_{t_1}^{\infty} \left(\frac{\partial}{\partial x_1} \bar{F}(x_1, t_2) \right)^\beta dx_1 = (\bar{F}(t_1, t_2))^{\beta-(\beta-1)} H_{11}^\beta(X, t_1, t_2) (\bar{F}(t_1, t_2))^\beta.$$

Differentiating with respect to t_1 and rearranging the terms we get

$$\begin{aligned} (\beta-1) \frac{\partial}{\partial t_1} H_{11}^\beta(X, t_1, t_2) - \beta(\beta-1) H_{11}^\beta(X, t_1, t_2) h_1(t_1, t_2) \\ + \beta h_1(t_1, t_2) - (h_1(t_1, t_2))^\beta = 0, \end{aligned}$$

where $h_1(t_1, t_2)$ is the vector-valued failure rate. Now using the argument given in Theorem 4.1 of Nanda and Paul (2006), it follows that $H_{11}^\beta(X, t_1, t_2)$ uniquely determines $h_1(t_1, t_2)$. Similarly $H_{12}^\beta(X, t_1, t_2)$ uniquely determines $h_2(t_1, t_2)$. Using the fact that the vector-valued failure rate determines the distribution uniquely (Johnson and Kotz (1975)) we infer that $H_1^\beta(X, t_1, t_2)$ uniquely determines $\bar{F}(t_1, t_2)$. This proves (1). To prove (2), we have, from

(2.10),

$$(\bar{F}(t_1, t_2))^\beta \exp\left((1 - \beta) H_{21}^\beta(X, t_1, t_2)\right) = \int_{t_1}^{\infty} \left(-\frac{\partial}{\partial x_1} \bar{F}(x_1, t_2)\right)^\beta dx_1.$$

Differentiating both sides with respect to t_1 we get

$$\begin{aligned} \beta h_1(t_1, t_2) \exp\left((1 - \beta) H_{21}^\beta(X, t_1, t_2)\right) - (h_1(t_1, t_2))^\beta &= (1 - \beta) \\ \frac{\partial}{\partial t_1} H_{11}^\beta(X, t_1, t_2) \exp\left((1 - \beta) H_{21}^\beta(X, t_1, t_2)\right). \end{aligned}$$

By using the same argument as in case 1, 2 follows immediately. $\square$

We now look into the situation where the vector-valued residual entropy is constant.

Theorem 3.2 Let $X = (X_1, X_2)$ be a non-negative, non-degenerate random vector admitting an absolutely continuous distribution function with respect to a Lebesgue measure and such that $H_i^\beta(X, t_1, t_2) < \infty$, $i = 1, 2$. The following relations

$$H_{1i}^\beta(X; t_1, t_2) = (c_1, c_2), \quad i = 1, 2 \quad (3.1)$$

$$H_{2i}^\beta(X; t_1, t_2) = (c_1, c_2), \quad i = 1, 2 \quad (3.2)$$

hold if and only if X is distributed as the bivariate exponential distribution with independent (exponential) marginals.

Proof When (3.1) holds with $i = 1$, using (2.10) we have

$$(1 - c_1(\beta - 1)) (\bar{F}(t_1, t_2))^\beta = \int_{t_1}^{\infty} \left(-\frac{\partial \bar{F}(x_1, t_2)}{\partial x_1}\right)^\beta dx_1.$$

Differentiating the above equation with respect to t_1 , and rearranging terms we get

$$h_1(t_1, t_2) = (c_1(\beta - 1) - 1)^{\frac{1}{\beta-1}} = \lambda_1 > 0.$$

Proceeding along similar lines using (2.12) we get

$$h_2(t_1, t_2) = (c_2(\beta - 1) - 1)^{\frac{1}{\beta-1}} = \lambda_2 > 0.$$

From Galambos and Kotz (1978) we have

$$\bar{F}(t_1, t_2) = \exp(-\lambda_1 t_1 - \lambda_2 t_2), \quad t_1, t_2 > 0 \quad (3.3)$$

as claimed.

Conversely when the distribution of X is specified by (3.3), by direct calculations we get

$$H_{1i}^\beta(X; t_1, t_2) = \frac{\beta - \lambda_i^{\beta-1}}{\beta(\beta - 1)}, \quad i = 1, 2$$

so that (3.1) holds. The proof for $H_2^\beta(X; t_1, t_2)$ is similar and hence omitted. $\square$

The following theorem provides characterizations for bivariate exponential, bivariate Pareto and bivariate beta distributions using a possible relationship between the vector-valued residual entropies and the vector-valued hazard rates.

Theorem 3.3 For the random vector X considered above, the relationship

$$H_{1i}^\beta(X, t_1, t_2) = \frac{1}{1 - \beta} \left(1 - k(h_i(t_1, t_2))^{\beta-1} \right), \quad i = 1, 2, \quad (3.4)$$

where $h_i(t_1, t_2)$ are the components of the bivariate failure rate, holds for all real $t_1, t_2 \geq 0$ if and only if X is distributed as

1. Gumbel's bivariate exponential distribution with survival function

$$\begin{aligned} \bar{F}(t_1, t_2) &= \exp(-\alpha_1 t_1 - \alpha_2 t_2 - \theta t_1 t_2), \\ \alpha_1, \alpha_2, t_1, t_2 &> 0, 0 < \theta < \alpha_1 \alpha_2 \end{aligned} \quad (3.5)$$

if $k = \frac{1}{\beta}$,

2. the bivariate Pareto distribution specified by

$$\begin{aligned} \bar{F}(t_1, t_2) &= (1 + a_1 t_1 + a_2 t_2 + b t_1 t_2)^{-c}, \\ t_1, t_2 &> 0, a_1, a_2, c > 0, 0 \leq b \leq (c + 1) a_1 a_2 \end{aligned} \quad (3.6)$$

if $k < \frac{1}{\beta}$; and

3. the bivariate Beta distribution specified by

$$\begin{aligned} \bar{F}(t_1, t_2) &= (1 - p_1 t_1 - p_2 t_2 + q t_1 t_2)^d, \\ 0 < t_1 < \frac{1}{p_1}, 0 < t_2 < \frac{1 - p_1 t_1}{p_2 - q t_1} \end{aligned} \quad (3.7)$$

if $k > \frac{1}{\beta}$.

Proof When (3.4) holds with $i = 1$, using (2.10) we can write

$$\int_{t_1}^{\infty} \left(-\frac{\partial \bar{F}(x_1, t_2)}{\partial x_1} \right)^{\beta} dx_1 = k (\bar{F}(t_1, t_2))^{\beta} (h_1(t_1, t_2))^{\beta-1}.$$

Differentiating the above equation with respect to t_1 on both sides, we get, after simplification,

$$(h_1(t_1, t_2))^{-2} \frac{\partial}{\partial t_1} h_1(t_1, t_2) = \frac{k\beta - 1}{k(\beta - 1)}.$$

This gives

$$h_1(t_1, t_2) = \frac{1}{\frac{(k\beta-1)}{k(\beta-1)} t_1 + c_1(t_2)}$$

where $c_1(t_2)$ is a constant independent of t_1 .

Proceeding along similar lines one can also get

$$h_2(t_1, t_2) = \frac{1}{\frac{(k\beta-1)}{k(\beta-1)} t_2 + c_2(t_1)}$$

where $c_2(t_1)$ is a constant independent of t_2 . The rest of the proof follows from Roy (1989). The “if” part of the theorem follows from Table 3.1 with similar expressions for $H_{12}^\beta(X, t_1, t_2)$ and $h_2(t_1, t_2)$. $\square$

Table 3.1 Expressions for $H_{11}^\beta(X, t_1, t_2)$ and $h_1(t_1, t_2)$

Distribution	$H_{11}^\beta(X, t_1, t_2)$	$h_1(t_1, t_2)$
(i) Exponential	$\frac{1}{1-\beta} \left(1 - \frac{(\alpha_1 + \theta t_2)^{\beta-1}}{\beta} \right)$	$\alpha_1 + \theta t_2$
(ii) Pareto	$\frac{1 - \left(\frac{c}{\beta(c+1)-1} \right) \left(\frac{c(a_1 + bt_2)}{1 + a_1 t_1 + a_2 t_2 + b t_1 t_2} \right)^{\beta-1}}{1 - \beta}$	$\frac{c(a_1 + bt_2)}{1 + a_1 t_1 + a_2 t_2 + b t_1 t_2}$
(iii) Beta	$\frac{1 - \frac{d}{(\beta(d-1)+1)} \left(\frac{d(p_1 - qt_2)}{1 - p_1 t_1 - p_2 t_2 + q t_1 t_2} \right)^{\beta-1}}{1 - \beta}$	$\frac{d(p_1 - qt_2)}{1 - p_1 t_1 - p_2 t_2 + q t_1 t_2}$

Theorem 3.4 For the random vector X considered in theorem 3.3 a relationship of the form

$$H_{2i}^\beta(X, t_1, t_2) = \frac{1}{1-\beta} \log k - \log h_i(t_1, t_2), \quad i = 1, 2, \quad (3.8)$$

where $h_i(t_1, t_2)$ are the components of the bivariate failure rate, holds for all real $t_1, t_2 \geq 0$ if and only if X follows any one of the three distributions specified by (3.5), (3.6) and (3.7) respectively according as $k = \frac{1}{\beta}$, $k < \frac{1}{\beta}$ and $k > \frac{1}{\beta}$.

Proof The proof of the “only if” part is similar to theorem 3.3 and hence omitted. The “if” part of the theorem follows from Table 3.2 with similar expressions

for $H_{22}^\beta(X, t_1, t_2)$ and $h_2(t_1, t_2)$. $\square$

Table 3.2 Expressions for $H_{21}^\beta(X, t_1, t_2)$ and $h_1(t_1, t_2)$

Distribution	$H_{21}^\beta(X, t_1, t_2)$	$h_1(t_1, t_2)$
(i) Exponential	$\frac{1}{1-\beta} \log \frac{1}{\beta} - \log(\alpha_1 + \theta t_2)$	$\alpha_1 + \theta t_2$
(ii) Pareto	$\frac{1}{1-\beta} \log \frac{c}{\beta(c+1)-1} - \log\left(\frac{c(a_1+bt_2)}{1+a_1t_1+a_2t_2+bt_1t_2}\right)$	$\frac{c(a_1+bt_2)}{1+a_1t_1+a_2t_2+bt_1t_2}$
(iii) Beta	$\frac{1}{1-\beta} \log \frac{d}{(\beta(d-1)+1)} - \log\left(\frac{d(p_1-qt_2)}{1-p_1t_1-p_2t_2+qt_1t_2}\right)$	$\frac{d(p_1-qt_2)}{1-p_1t_1-p_2t_2+qt_1t_2}$

4. Numerical illustration

To illustrate and contrast our notions we calculate the vector-valued β -residual entropy for the data of US income and wealth reported by Anderson (1999). For the data of income and wealth, we fit the bivariate exponential distribution with estimated parameters

$$\lambda_1 = 0.00001114 \text{ and } \lambda_2 = 0.00002788.$$

The correlation coefficient between income and wealth is $\rho = 0.837$. Using the well-known relationship

$$\rho = \frac{\theta}{\lambda_1 + \lambda_2 + \theta}$$

we get the value of $\theta = 0.0002394$. The minimum values of income and wealth are reported in the data as 17741 and 19136, respectively. We truncated the data and approximated the truncation points to $t_1 = t_2 = 25,000$. Using (2.5) and (2.7) the vector-valued β -residual entropy of the bivariate exponential

distribution is simplified as

$$H_1^\beta(X, t_1, t_2) = (-1.2619, -1.26191).$$

Using (1.3), the univariate β -residual entropy for income is calculated as

$$H_1^\beta(X, t) = 1.99555.$$

Similarly, the univariate β -residual entropy for wealth is given as

$$H_1^\beta(X, t) = 1.99296.$$

Using (2.10) and (2.12) the second kind of vector-valued β -residual entropy of the bivariate exponential distribution is simplified as

$$H_2^\beta(X, t_1, t_2) = (-0.978328, -0.978331).$$

Using (1.4), the univariate β -residual entropy for income is calculated as

$$H_2^\beta(X, t) = -12.2159.$$

Similarly, the univariate β -residual entropy for wealth is given as

$$H_2^\beta(X, t) = -11.2985.$$

If $X = (X_1, X_2)$ is a bivariate random vector representing the income and wealth, with X_1 representing the income and X_2 representing the wealth, then $H_{11}^\beta(X, t_1, t_2)$ measures the expected uncertainty contained in the distribution of the amount of income subject to the condition that the wealth of the population exceeds a threshold t_2 . Similarly $H_{12}^\beta(X, t_1, t_2)$ measures the expected uncertainty contained in the distribution of the amount of wealth subject to the condition that the income of the population exceeds a threshold t_1 . A similar interpretation can also be given for $H_2^\beta(X, t)$.

It may be observed from the above example that the uncertainty reduces when we consider the two attributes, viz. income and wealth, jointly. This is an indication that, in an economy, when we consider one particular attribute

it may seem to be more unstable, whereas when it is considered along with other attributes, the overall economy may not be that unstable. We therefore recommend considering multivariate residual entropy to measure uncertainty in such situations.

5. Generalization to higher dimensions

In this section we generalize the β -residual entropies defined in (1.3) and (1.4) to higher dimensions. Let $X = (X_1, X_2, \dots, X_p)$ be an absolutely continuous random vector with density function $f(x_1, x_2, \dots, x_p)$. The survival function is

$$\bar{F}(t_1, t_2, \dots, t_p) = P(X_1 > t_1, X_2 > t_2, \dots, X_p > t_p).$$

Let $Y_i = X_i | X_1 > t_1, X_2 > t_2, \dots, X_{(i-1)} > t_{(i-1)}, X_{(i+1)} > t_{(i+1)}, \dots, X_p > t_p, i = 1, 2, \dots, p$ represent a random variable with density function $f(x_i | X_1 > t_1, X_2 > t_2, \dots, X_p > t_p)$ and survival function $\bar{F}(x_i | X_1 > t_1, X_2 > t_2, \dots, X_p > t_p)$.

Noting that

$$\bar{F}(t_1, t_2, \dots, t_p) = \bar{F}(t_1 | X_2 > t_2, \dots, X_p > t_p) P(X_2 > t_2, \dots, X_p > t_p),$$

we get

$$\frac{\partial}{\partial t_1} \bar{F}(t_1, t_2, \dots, t_p) = -f(t_1 | X_2 > t_2, \dots, X_p > t_p) P(X_2 > t_2, \dots, X_p > t_p).$$

This gives

$$f(t_1 | X_2 > t_2, \dots, X_p > t_p) = \frac{-1}{P(X_2 > t_2, \dots, X_p > t_p)} \frac{\partial}{\partial t_1} \bar{F}(t_1, t_2, \dots, t_p).$$

We define the first kind of residual entropy of order β for the random variable $Y_i, i = 1, 2, \dots, p$ as

$$H_{1i}^\beta(X, t) = \frac{1}{1-\beta} \left(1 - \int_{t_i}^{\infty} \left(\frac{f(x_i | X_1 > t_1, \dots, X_p > t_p)}{\bar{F}(t_i | X_1 > t_1, \dots, X_p > t_p)} \right)^\beta dx_i \right). \quad (4.1)$$

(4.1) can also be written as

$$H_{1i}^{\beta}(X, t) = \frac{1}{1 - \beta} \left(1 - \int_{t_i}^{\infty} \left(\frac{-\frac{\partial}{\partial x_i} \bar{F}(x_i, t_2, \dots, t_p)}{\bar{F}(t_1, t_2, \dots, t_p)} \right)^{\beta} dx_i \right). \quad (4.2)$$

Analogous to the definition of the vector-valued failure rate we give an alternative definition for the first kind of residual entropy of order β :

Definition 4.1

For a non-negative random vector $X = (X_1, X_2, \dots, X_p)$ admitting an absolutely continuous distribution function, the β -residual entropy vector is defined as

$$H_1^{\beta}(X, t) = (H_{11}^{\beta}(X, t), H_{12}^{\beta}(X, t), \dots, H_{1p}^{\beta}(X, t)), \quad (4.3)$$

where $H_{1i}^{\beta}(X, t)$, $i = 1, 2, \dots, p$ are defined as in (4.1).

Similarly, the second kind of β -residual entropy for the random variable Y_i , $i = 1, 2, \dots, p$ is defined as

$$H_{2i}^{\beta}(X, t) = \frac{1}{1 - \beta} \log \left(\int_{t_i}^{\infty} \left(\frac{f(x_i | X_1 > t_1, \dots, X_p > t_p)}{\bar{F}(t_i | X_1 > t_1, \dots, X_p > t_p)} \right)^{\beta} dx_i \right). \quad (4.4)$$

(4.4) can also be written as

$$H_{2i}^{\beta}(X, t) = \frac{1}{1 - \beta} \log \left(\int_{t_i}^{\infty} \left(\frac{-\frac{\partial}{\partial x_i} \bar{F}(x_i, t_2, \dots, t_p)}{\bar{F}(t_1, t_2, \dots, t_p)} \right)^{\beta} dx_i \right). \quad (4.5)$$

Definition 4.2

For a non-negative random vector $X = (X_1, X_2, \dots, X_p)$ admitting an absolutely continuous distribution function, the β -residual entropy vector of the second kind is defined as

$$H_2^{\beta}(X, t) = (H_{21}^{\beta}(X, t), H_{22}^{\beta}(X, t), \dots, H_{2p}^{\beta}(X, t)), \quad (4.6)$$

where $H_{2i}^\beta(X, t)$, $i = 1, 2, \dots, p$ is given by (4.4) .

Generalization shall be obtained for the results in Section 3 in the p -dimensional set-up. As the results are fairly straightforward, they are not pursued further.

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